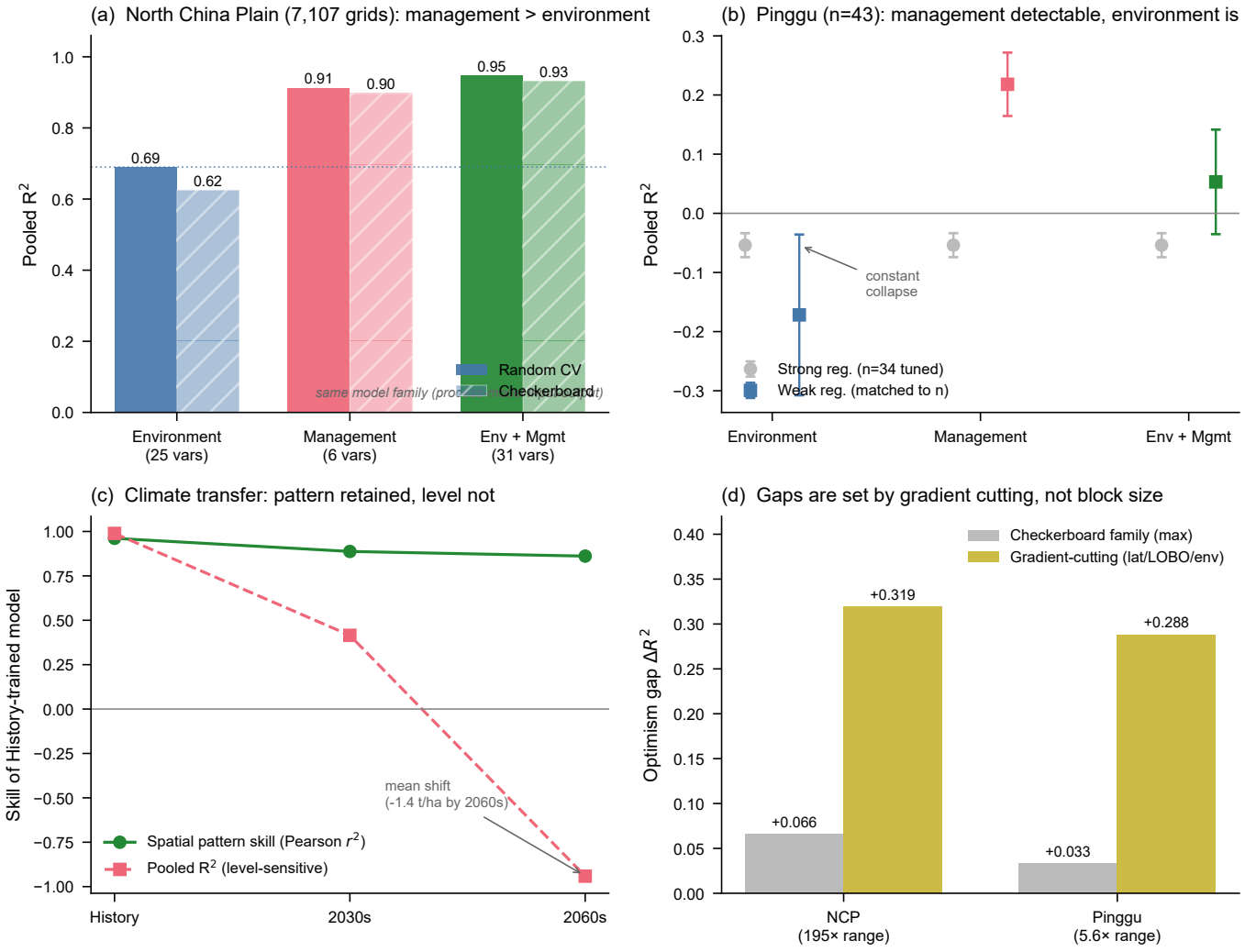

(a) North China Plain (7,107 grids): management > environment
Pooled R²
1.0
0.8
0.6
0.4
0.2
0.0
0.69
0.62
0.91
0.90
0.95
0.93
Random CV
Checkerboard
same model family (prot
Environment
(25 vars)
Management
(6 vars)
Env + Mgmt
(31 vars)
(b) Pinggu (n=43): management detectable, environment is n
Pooled R²
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
constant
collapse
Strong reg. (n=34 tuned)
Weak reg. (matched to n)
Environment
Management
Env + Mgmt
(c) Climate transfer: pattern retained, level not
Skill of History-trained model
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
mean shift
(-1.4 t/ha by 2060s)
Spatial pattern skill (Pearson r²)
Pooled R² (level-sensitive)
History
2030s
2060s
(d) Gaps are set by gradient cutting, not block size
Optimism gap ΔR²
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Checkerboard family (max)
Gradient-cutting (lat/LOBO/env)
+0.066
+0.319
+0.033
+0.288
NCP
(195× range)
Pinggu
(5.6× range)